CTC BESPOKE XML SCHEMA

VERSION 1.6

1. Document Control

Date	Updated By	Detail of change
18/08/2020	CTC	<agent_name> made optional <fee_minimum>, <fee_maximum>, <fee_fund_basis>, <fund_ref>, <fee_increases> added to <product_fee> Changes to <withdrawals>
03/09/2020	CTC	<fee_minimum>, <fee_maximum> replaced with <fee_constraints><maximum_value>/<minimum_value> <fund_name> added to <fund_split> <stream_applicable> added to <adviser_fee>
08/09/2020	CTC	<stream_applicable> on <adviser_fee> updated to clarify not mandatory for ongoing adviser fees Optional ‘type' attribute added to <fund_split> <allocation_basis> and <allocation> added to 
01/10/2020	CTC	<fee_name> added to <adviser_fee>
05/10/2020	CTC	<initial_charge> added to <fund_split>
01/12/2020	CTC	Correction for fund specific tax free cash - each <fund_ref>/<split_per> should be contained within <fund_split>
22/02/2021	CTC	Correction for crystallisation and withdrawal frequency options

2. XML Interface to Calculation Engine

The XML message has the following outline structure :

<message>
<header>
<content>
<quotations>
<investor>
<quotation>
<plan_stream>
<premium_split>
<withdrawals>

<product_fees>
<adviser_fees>
<annuity_basis>
</quotation>
</quotations>
<results>
</content>
</message>

Header Contents

Request Message

The request should appear as

<header>
<message_version>CTC1</message_version>
<created_date>yyyy-MM-dd HH:mm:ss</created_date>
<mode>Request</state>
</header>

Response Message – Success

<header>
<created _date>yyyy-MM-dd HH:mm:ss</created_date>
<response_date>yyyy-MM-dd HH:mm:ss</response_date>
<mode>Response</state>
<response>SUCCESS</response>
</header>

Response Message – Error

<header>
<created _date>yyyy-MM-dd HH:mm:ss</created_date>
<response_date>yyyy-MM-dd HH:mm:ss</response_date>
<mode>Response</state>
<response>ERROR</response>
<errors>
<error>DESCRIPTION</error>
</errors>
</header>

The <errror> desciptions are as follows:

Description	**Meaning**
Unknown Product Code	There is an invalid product code in the XML
Error reading XML message	Improperly formed XML message
An error has occurred when trying to produce the output	There has been a system calculation/output product error
Specified fund not recognised.	Produced where a fund reference is not recognised. The reference is shown after the message (eg ‘Specified fund not recognised. LU0278930234, ,’).

a. quotations

This is a non repeating group.

Node	Attributes	Element Description	Mandatory/ Optional
<quotations>			
<effective_date>		Effective date of quotation (yyyy-mm-dd)	M
<output_required>		This will have the following values: “Yes” “No”	M
<output_reference>		Identifying reference to be included when naming PDF output generated by the message. (max 45 characters)	O
<agent_name>		Name of agent for inclusion in output (max 130 characters)	O
<client_type>		Values are: “Company” “Trust” “Individual”	O
<name>		Name of company or trust. This is only applicable if client_type = “Company” or “Trust”	O

Example 1

Illustrations are for an individual:

```
<quotations>
    <effective_date>2015-11-24</effective_date>
    <output_required>Yes</output_required>
    <output_reference>ABCD1234</output_reference>
    <agent_name>XYZ Financial Advisers Limited</agent_name>
    <client_type>Individual</client_type>
```

Example 2

Illustrations are for a company:

```
<quotations>
        <effective_date>2015-11-24</effective_date>
        <output_required>Yes</output_required>
        <output_reference>ABCD1234</output_reference>
        <agent_name>XYZ Financial Advisers Limited</agent_name>
        <client_type>Company</client_type>
        <name>Example Company Ltd</name>
```

b. investor

This a repeating group

Node	Attributes	Element Description	Mandatory/ Optional
<investor>		Clients included in the illustration	M
<title>		Title	O
<forename>		Forenames (max 100 characters)	O
<surname>		Surname (max 50 characters)	O
<sex>		The values here are “Male” or “Female”	O
<life_type>		Values are: “Applicant” “Life Assured” “Both” “Dependant” “None”	O
<marital_status>		Values are: “Civil Partnership” “Divorced” “Married” “Partnered” “Single” “Widowed” “Unknown”	O
<date_of_birth>		Date of birth (yyyy-mm-dd)	O
<exit_age>		Planned retirement age/conversion age for pension products	O

Example 1

Illustration including pension product for individual with no spouse’s details:

```
<investor>
        <title>Ms</title>
        <forename>Main Life</forename>
        <surname>Example</surname>
        <sex>Female</sex>
        <life_type>None</life_type>
        <marital_status>Unknown</marital_status>
        <date_of_birth>1955-06-13</date_of_birth>
        <exit_age>73</exit_age>
</investor>
```

Example 2

Illustration including pension product for individual with spouse's details:

```
<investor>
        <title>Mrs</title>
        <forename>Main Life</forename>
        <surname>Example</surname>
        <sex>Female</sex>
        <life_type>None</life_type>
        <marital_status>Married</marital_status>
        <date_of_birth>1955-06-13</date_of_birth>
        <exit_age>73</exit_age>
</investor>
<investor>
        <title>Mr</title>
        <forename>Spouse</forename>
        <surname>Example</surname>
        <sex>Male</sex>
        <life_type>Dependant</life_type>
        <marital_status>Married</marital_status>
        <date_of_birth>1964-08-31</date_of_birth>
</investor>
```

Example 3

Investment Bond with multiple lives:

```
<investor>
        <title>Mrs</title>
        <forename>Main Life</forename>
        <surname>Example</surname>
        <sex>Female</sex>
        <life_type>Life Assured</life_type>
        <date_of_birth>1955-06-13</date_of_birth>
</investor>
<investor>
        <title>Mr</title>
```

```
        <forename>Bond Life One</forename>
        <surname>Example</surname>
        <sex>Male</sex>
        <life_type>Applicant</life_type>
        <date_of_birth>1964-08-31</date_of_birth>
</investor>
<investor>
        <title>Mr</title>
        <forename>Bond Life Two</forename>
        <surname>Example</surname>
        <sex>Male</sex>
        <life_type>Both</life_type>
        <date_of_birth>1995-01-14</date_of_birth>
</investor>
```

c. quotation

This is a repeating group.

Node	Attributes	Element Description	Mandatory/ Optional
<quotation>		Quotation group repeated for each product in the quote	M
	product_ref product_type	The reference of the product that the projection is based on. 'Self Invested Personal Pension'	M M
<quotation_type>		Refers to the pension element of the message. "Accumulation" "Drawdown" "SMPI" NB Use "Accumulation" for all non-pension products.	M
<inflation_adjustment_indicator>		Specifies whether inflation adjustment should be applied – 'Yes' or 'No'	O
<current_max_income>		The current GAD maximum income for capped drawdown illustrations	O
<next_review_date>		The date of next 3 yearly review (yyyy-mm-dd) for capped drawdown illustrations	O
<inc_already_taken>		Amount of income already taken in the current review year for capped drawdown illustrations.	O

Example 1

Flexi-access Drawdown illustration for a Self Invested Personal Pension product:

```
<quotation product_ref="SIPPEN">
        <quotation_type>Flexi Access Drawdown</quotation_type>
</quotation>
```

Example 2

Illustration for an ISA product:

```
<quotation product_ref="ISACC">
        <quotation_type>Accumulation</quotation_type>
</quotation>
```

d. plan_stream

This will repeat for each payment type in respect of the individual product.

Node	Attributes	Element Description	Mandatory /Optional
<plan_stream>			M
	plan_stream_key	Unique stream key within the request message e.g. 'STREAM1'	M
<contribution_type>		"Member" "Employer" "Transfer Value" "Existing Fund Value"	M
<net_gross>		Use for <contribution_type> of 'Member' – 'Net' or 'Gross'	M
<amount>		Amount of total contribution	M
<frequency>		This can have the following values: "Weekly" "Monthly" "Quarterly" "Half Yearly" "Annually" "Single" "Transfer Value" and "Existing Fund Value" must always be "Single"	M
<crystallised>		Is this stream crystallised. The values are: "Yes" "No" Only "Transfer Value" and "Existing Fund Value" can be "Yes" or "No". All other payment types must be "No".	M
<plan_stream_start>		Used to specify when the stream should start	O
<start_type>		"Specified Date"	

<start_date> <plan_stream_end> <end_type> <end_date>		The date the stream should start Used to specify when the stream should end “Specified Date” The date the stream should end	O

Example 1

Illustration with existing fund values (crystallised and non-crystallised) and a new member single payment:

```
<plan_stream plan_stream_key="100000">
        <contribution_type>Existing Fund Value</contribution_type>
        <amount>543210.98</amount>
        <frequency>Single</frequency>
        <crystallised>No</crystallised>
</plan_stream>
<plan_stream plan_stream_key="100001">
        <contribution_type>Existing Fund Value</contribution_type>
        <amount>234561.66</amount>
        <frequency>Single</frequency>
        <crystallised>Yes</crystallised>
</plan_stream>
<plan_stream plan_stream_key="ABCDE">
        <contribution_type>Member</contribution_type>
        <amount>12345.67</amount>
        <frequency>Single</frequency>
        <crystallised>No</crystallised>
</plan_stream>
```

Example 2

Illustration with member and employer regular contributions:

```
<plan_stream plan_stream_key="ABC100">
        <contribution_type>Member</contribution_type>
        <amount>650.52</amount>
        <frequency>Quarterly</frequency>
        <crystallised>No</crystallised>
</plan_stream>
<plan_stream plan_stream_key="ABC102">
        <contribution_type>Employer</contribution_type>
        <amount>888.01</amount>
        <frequency>Quarterly</frequency>
        <crystallised>No</crystallised>
```

</plan_stream>

e. premium_split

This is a repeating group

Node	Attributes	Element Description	Mandatory/ Optional
<premium_split>		Premium split group	O
<stream_applicable>		Can be repeated for each stream key that applies to the investment	M
	plan_stream_key	Unique stream key linked to <plan_stream>	M
<fund_split>	type	Attribute optional – use 'COMMPROP' for Commercial Property	M
<fund_ref>		Fund Reference code	M
<fund_name>		This is the name of the fund/investment	
<split_per>		Percentage of total contribution invested in the fund	M – one of <split_per> and <split_amount> is M, only one type should be used to define the total <premium_split>
<split_amount>		Amount invested in the fund	M
<rate_of_return>		The mid growth rate that applies for the fund	O
<management_charge>		Any % management charge that should apply for the fund	O
<initial_charge>		% initial charge on the fund	O
<generic_fund>		Values of 'Yes' and 'No' - set to 'Yes' if the <fund_split> contains <rate_of_return> and/or <management_charge>	O

Example

Example where there are two payments under <plan_stream> (plan_stream_keys = “A100” and “B200”) and both are to have the same investment split applied to them.

```
<premium_split>
<stream_applicable plan_stream_key = "A100"/>
<stream_applicable plan_stream_key = "B200"/>
        <fund_split>
                <fund_ref> LU0651906728</fund_ref>
                <split_per>60</split_per>
        </fund_split>
        <fund_split>
                <fund_ref> LU0231455709</fund_ref>
                <split_per >40</split_per >
        </fund_split>
</premium_split>
```

f. withdrawals and crystallisations

This is a repeating group.

Node	Attributes	Element Description	Mandatory/ Optional
<withdrawals>		Withdrawals	O
<basis>		Value should be: “Fixed” “Maximum GAD” “Percentage of Maximum GAD”	O
<amount>		Amount of withdrawal per frequency. Should only appear if basis = “Fixed” or for a non-pension product	O
<percent>		This is used for when basis = “Percentage of Maximum GAD”	O
<frequency>		Withdrawal frequency. The values are: “Single” “Monthly” “Quarterly” “Half Yearly” “Annually” “2 Months” “4 Months”	O
<start_date>		Date fromwhich regular withdrawals are to commence (yyyy-mm-dd)	O
<increases>		Specifies the rate of increase of the withdrawal	O
<percent>		The percentage rate of increase	
<drawdown_type>		‘Flexible', 'Capped', 'UFPLS'	M

This is a repeating group.

Node	Attributes	Element Description	Mandatory/ Optional
<crystallisation>		Crystallisation and tax free cash instruction	O
<drawdown_type>		‘Flexible’ or ‘Capped’	M

<amount>		The amount being crystallised	O
<frequency>	type	The frequency of crystallisation – 'Single', 'Monthly', 'Quarterly', 'Half Yearly', 'Yearly'	O
		This field is only applicable for Drawdown and denotes the Tax Free Cash to be taken.	
		Tax Free Cash type. Values are: "Maximum" "Amount" "Percentage"	O
<amount>		If the Tax Free Cash is an "Amount"	O
<percent>		If the Tax Free Cash is a "Percentage" of fund, the percentage is entered here.	
<allocation_basis>		Use to specify fund specific Tax Free Cash – value of 'Specified'	O
<allocation>		Holds the funds that the Tax Free Cash is to be deducted from	
<fund_split>		Reference of the fund	
<fund_ref>		The percentage of the Tax Free Cash to be deducted from the fund	
<split_per>			
		Used for tax free cash as income	O
<phasedcash>		The amount of tax free cash as income	O
<amount>		Used with <phasedcash> - set to 'Non taxable income'	O
<phased_planning_option>			

g. product_fees

This group will appear once and contains the product_fees details for the individual product.

Where the section is present in the message it must **appear for each product**and contain **all the charges** applicable (ie the standard charges in the system standing data will be ignored entirely).

As noted below; tiered charges should be provided for each product in the quotation and contain **identical** data.

Node	Attribute	Element Description	
<product_fees>		Product fees (non-tiered) group Where this section is present; all non-tiered fees must be specified here (both default and non-default).	O
<product_fee>		Repeat for every non-tiered fee applicable to the product.	O
<fee_name>		The name that will appear in output to describe the fee (max 50 characters)	M
<fee_basis>		This will have the following values: “Single” “Ongoing”	M
<fee_frequency>		 “Single” “Monthly” “Quarterly” “Half Yearly” “Yearly”	M
<fee_timing>		This will have the following values: “In Advance” “In Arrears” “Before Growth” “After Growth”	M

<fee_value_rate>		“Per Frequency” or “Per Annum”	M
<fee_increases>		Specifies any increases for the fee	O
<increase_type>		‘Fixed'	
<percentage>		The percentage rate of increase	
<fee_constraints>			O
<maximum_value>		The maximum amount that should be charged	O
<minimum_value>		The minimum amount that should be charged	O
<fee_tier_selection>		Must always have the following value: “Each Threshold”	O (M if tiered charge)
<fee_tiers>		Should appear once for each product in the quotation with the same values each time.	O
	type	This will have the following values: “Percentage” “Amount”	O
<tier>	upperval	This is the upper level at which the tier applies. Where there is no maximum (ie because the top tier has been reached); a value of “999999999” should be provided.	O
<fee_fund_basis>		This will have the following values: ‘Pro-Rata’ or ‘Client Select’	O
<fund_split>		Repeat for each fund relevant to the ‘Client Select’ option	M if <fee_fund_basis> is ‘Client Select’
<fund_ref>		The reference of the fund	M if <fee_fund_

		basis> is ‘Client Select’
<fee_start>		M
<start_type>	“Immediately” ‘Specified Date’	M
<start_date>	Used if <start_type> is ‘Specified Date'	M
<fee_end>		M
<end_type>	“Maturity” 'Specified Date'	M
<end_date>	Used if <end_type> is ‘Specified Date'	M
<amount>	Required for non-tiered charges. The charge should be set to one of amount or percentage	O
<percentage>	Required for non-tiered charges.The percentage rate is an annual rate	O
<plus_vat>	Should VAT be applied to the fee? This will have the following values: “Yes” “No”	O

Example

Product with the following charges:

- Tiered Charges of:

Tier	**Charge**
£0.00 - £999,999.99	0.35%
£1,000,000.00 - £2,999,999.99	0.29%
£3,000,000.00 - £4,999,999.99	0.24%
£5,000,000.00 +	0.20%

- Ongoing fee of 0.33% of fund payable monthly in arrears.
- Initial one-off fee of £100.00.

<productfees>
    <product_fee>

```
        <fee_name>Tiered Annual Platform Charge</fee_name>
        <fee_basis>Ongoing</fee_basis>
        <fee_frequency>Monthly</fee_frequency>
        <fee_timing>In Arrears</fee_timing>

        <fee_value_rate>Per Annum</fee_value_rate>
        <fee_tier_selection>Each Threshold</fee_tier_selection>
        <fee_tiers type="Percentage">
                <tier upperval="999999.99">0.35</tier>
                <tier upperval="2999999.99">0.29</tier>
                <tier upperval="4999999.99">0.24</tier>
                <tier upperval="999999999">0.2</tier>
        </fee_tiers>
        <fee_start>
                <start_type>Immediately</start_type>
        </fee_start>
        <fee_end>
                <end_type>Maturity</end_type>
        </fee_end>
        <plus_vat>No</plus_vat>
</product_fee>
<product_fee>
        <fee_name>Bond Charge</fee_name>
        <fee_basis>Ongoing</fee_basis>
        <fee_frequency>Monthly</fee_frequency>
        <fee_timing>In Arrears</fee_frequency>
        <fee_value_rate>Per Annum</fee_value_rate>
        <fee_start>
                <start_type>Immediately</start_type>
        </fee_start>
        <fee_end>
                <end_type>Maturity</end_type>
        </fee_end>
        <percentage>0.33</percentage>
        <plus_vat>No</plus_vat>
</product_fee>
<product_fee>
        <fee_name>Account Opening Charge - OPB</fee_name>
        <fee_basis>Single</fee_basis>
        <fee_frequency>Single</fee_frequency>
        <fee_timing>In Advance</fee_frequency>
        <fee_value_rate>Per Annum</fee_value_rate>
        <fee_start>
                <start_type>Immediately</start_type>
        </fee_start>
        <fee_end>
                <end_type>Maturity</end_type>
        </fee_end>
        <amount>100</amount>
        <plus_vat>No</plus_vat>
```

```
        </product_fee>
</product_fees>
```

h. adviser_fees

This group will appear once and contains the adviser_fees details for each pot of money.

Node	Attribute	Element Description	
<adviser_fees>		Adviser fees group	O
<adviser_fee>		Can be repeated, if necessary	O
<stream_applicable>	Plan_stream_key	Repeat for each stream that the fee should be attached to	M if <fee_basis> is 'Initial'
<fee_name>		Use if a specific name for the fee is needed for the output document	O
<fee_basis>		This will have the following values "Initial" "Ongoing"	M
<fee_frequency>		"Single" "Monthly" 'Quarterly' 'Half Yearly' 'Yearly'	M
<fee_timing>		'In Advance', 'Before Growth', 'After Growth', "In Arrears"	M
<fee_value_rate>		'Per Annum' or 'Per Frequency'	M
<fee_tier_selection>		Must always have the following value: "Each Threshold"	O (M if tiered charge)
<fee_tiers>		Should appear once for each product in the quotation with the same values each time.	O
	type	This will have the following values: "Percentage" "Amount"	O

<tier>	upperval	This is the upper level at which the tier applies. Where there is no maximum (ie because the top tier has been reached); a value of “999999999” should be provided.	O
<fee_start>			M
<start_type>		“Immediately” ‘Specified Date’	M
<start_date>		Used if <start_type> is ‘Specified Date'	M
<fee_end>			M
<end_type>		“Maturity” 'Specified Date'	M
<end_date>		Used if <end_type> is ‘Specified Date'	M
<amount>		The charge should be set to one of amount or percentage	O
<percentage>		The percentage rate is an annual rate	O
<plus_vat>		This will have the following values: “Yes” “No”	O

Example

Both intial and ongoing charges apply to the product:

```
<adviser_fees>
        <adviser_fee>
                <stream_applicable plan_stream_key=”CONT1”/>
                <fee_basis>Initial</fee_basis>
                <fee_frequency>Single</fee_frequency>
                <fee_timing>In Arrears</fee_timing>
                <fee_value_rate>Per Annum</fee_value_rate>
                <fee_start>
                        <start_type>Immediately</start_type>
                </fee_start>
                <fee_end>
                        <end_type>Maturity</end_type>
                </fee_end>
                <amount>500.00</amount>
        </adviser_fee>
```

```
        <adviser_fee>
                <stream_applicable plan_stream_key=”CONT1”/>
                <fee_basis>Ongoing</fee_basis>
                <fee_frequency>Monthly</fee_frequency>
                <fee_timing>In Arrears</fee_timing>
                <fee_value_rate>Per Annum</fee_value_rate>
                <fee_start>
                        <start_type>Immediately</start_type>
                </fee_start>
                <fee_end>
                        <end_type>Maturity</end_type>
                </fee_end>
                <percentage>0.75</percentage>
                <plus_vat>Yes</plus_vat>
        </adviser_fee>
</adviser_fees>
```

i. annuity_basis

This is a non-repeating group.

Node	Attributes	Element Description	Mandatory/ Optional
<annuity_basis>		Annuity group	M If the quotation is for a Pension product otherwise O
<annuity_frequency>		The frequency at which the pension is to be paid. This will have one of the following values: “Monthly” “Quarterly” “Half Yearly” “Annually”	O
<annuity_timing>		This can have one of the following values: “Advance” “Arrears”	O
<escalation>		This can have one of the following: “Fixed” “Level” “LPI” “RPI”	O
<percent>		This is only used if escalation = “Fixed” and will have the % of pension increase in payment.	O
<guaranteed_period>		The guarantee period	O
<spouse_percentage>		The spouse’s percentage	O

Example 1

Quarterly annuity with 3% pa increases in payment and a 50% spouse’s pension:

```
<annuity>
        <annuity_frequency>Quarterly</annuity_frequency>
        <annuity_timing>Advance</annuity_timing>
        <escalation>Fixed</escalation>
        <percent>3</percent>
        <spouse_percentage>50</spouse_percentage>
</annuity>
```

Example 2

Monthly annuity with LPI increases in payment, 5 year guarantee and no spouse's pension:

```
<annuity>
        <annuity_frequency>Monthly</annuity_frequency>
        <annuity_timing>Advance</annuity_timing>
        <escalation>LPI</escalation>
        <guaranteed_period>5</guaranteed_period>
</annuity>
```

j. results (Response)

This is an additional response message which send the path and name of the assembled PDF output file.

Node	Attributes	Element Description
<document_location>		Name of the PDF files that has been generated.